

KRANTHI KUMAR MUPPALLA

Design & Verification Engineer

Bangalore, India kranthimuppalla2@gmail.com +91 7330853702

SUMMARY

Motivated and detail-oriented VLSI Design and Verification Engineer with a solid foundation in Digital Design, Verilog, SystemVerilog, and UVM. Hands-on experience in writing testbenches, developing UVM-based environments, and verifying ASIC/FPGA modules. Strong understanding of AXI/APB/AHB protocols, SystemVerilog Assertions, and simulation tools such as QuestaSim, VCS, and ModelSim. Enthusiastic about contributing to cutting-edge semiconductor projects in collaborative environments.

EDUCATION

Bachelor of Technology in Electrical and Electronics Engineering

July 2020 – May 2024

Audisankara Institute of Technology, Gudur

TECHNICAL SKILLS

- **Languages:** Verilog, System Verilog
- **Verification:** UVM, Functional Coverage, Assertions (SVA)
- **Protocols:** AXI, AHB, APB
- **Simulation Tools:** QuestaSim, ModelSim, VCS
- **Debug Tools:** DVE, SimVision
- **Scripting:** Shell, Python, TCL (basic)
- **Concepts:** FSM, FIFO, Memory, Digital Logic, OOPs

EXPERIENCE

VLSI Design & Verification Internship – VLSI Guru Institute, Bangalore

June 2024 – May 2025

- Gained hands-on experience in RTL design and verification using Verilog and SystemVerilog.
- Developed UVM-based environments and automated simulation tasks using scripting.
- Performed functional and code coverage analysis, wrote assertions, and debugged simulations.
- Worked on real-time verification of protocols and IP-level designs.

PROJECTS

- **Memory Design and Verification**
 - Designed a synchronous memory module using Verilog.
 - Developed testbench and verified read/write operations with multiple scenarios.
 - Performed functional simulation using QuestaSim
- **FIFO (First-In-First-Out) Design and Verification**
 - Implemented synchronous and asynchronous FIFO in Verilog.
 - Verified overflow, underflow, and reset corner cases.
 - Applied coverage-driven verification with SVA.
- **Interrupt Controller Design and Verification**
 - Designed controller supporting multiple interrupts with priority handling.
 - Built UVM-based testbench for edge/level triggering scenarios.
 - Debugged using ModelSim/QuestaSim
- **SPI Protocol Design and Verification**
 - Implemented SPI Master and Slave as per protocol specs.
 - Verified with CPOL and CPHA variations.
 - Added assertions to confirm protocol correctness.

- **Pattern Detector Design and Verification**
 - Designed pattern detector to identify bit sequences.
 - Verified detection across all pattern combinations.
 - Applied functional coverage for completeness.

CERTIFICATIONS

Digital Design & Verification – VLSI Guru Institute

UVM Methodology & RTL Verification – Internal Projects

SOFT SKILLS

Problem-Solving | Communication | Adaptability | Teamwork | Quick Learner | Debugging